

NetST — User Manual & Software Guide

English |

NetST — Haplotype Network Analysis Tool / Audience: researchers in population genetics, phylogeography, and molecular epidemiology

NetST covers sequence import, standardization, multiple-sequence alignment, haplotype identification, network construction, interactive visualization, and interpretation. This manual follows the current source code and README.md.

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1. Overview

NetST is an open-source desktop application for project-based haplotype network analysis with metadata integration. It covers the path from sequence import to interactive visualization while retaining the input data, sample-to-haplotype mapping, categorical and continuous metadata, construction settings, analytical results, and visualization state as one reusable project. Established bioinformatics engines (MAFFT, MUSCLE, fastHaN, and McAN) and the NetST RMST reimplementations are accessed through one graphical interface. Long-running tasks run on background threads and can be cancelled.

Core capabilities

- **Multi-format import:** FASTA, NEXUS, PHYLIP, VCF/VCF.GZ; metadata can be parsed from sample IDs or imported separately from CSV/TSV.
- **Multiple network algorithms:** Original TCS, Modified TCS, MSN, MJN (fastHaN), built-in RMST, and McAN minimum-cost arborescence network.
- **Multi-trait concentric-layer visualization:** an enhanced tcsBU / D3.js view displays several categorical and continuous traits on the same network with a matching legend; export to SVG/PNG/JPG/PDF.
- **Persistent project state:** a portable .netst.json file records inputs, sample-to-haplotype membership, method settings, metadata mappings, analysis outputs, and the current network view for later replay.
- **Interpretation module:** visual quality, diversity, genetic-distance/PCoA, and network-topology metrics that help you understand the data and network at a glance.
- **Bilingual UI:** menus, tabs, table headers, logs, and result summaries refresh together on a language switch.

Design principle: topological description and biological inference are strictly separated. Centrality, bridges, PCoA proximity, etc. are **exploratory descriptors**; the software never automatically equates them with ancestry, origin, transmission source, or true populations.

2. Workflow at a Glance

FASTA / NEXUS / PHYLIP / VCF sequences



MAFFT or MUSCLE multiple-sequence alignment

Identify unique haplotypes → PHYLIP / CSV / FASTA files

fastHaN (TCS / MSN / MJN), built-in RMST, or McAN directed network

Convert to a common GML and generate tcsBU config

Interactive visualization in Qt WebEngine — optional: Interpretation Analysis

Portable project export / replay (‘<project>.netst.json’)

Stage	Module	Notes
Import & preprocess	File menu + standardization dialog	Four sequence formats + separate metadata table; clean/split sample IDs at import
Alignment	MAFFT / MUSCLE	Runs automatically during network

Stage	Module	Notes
		construction; can also be run alone under Tools
Haplotype identification	built-in	Collapses identical aligned sequences into unique haplotypes (H1, H2, ...)
Network construction	fastHaN / RMST / McAN	Six algorithms, unified GML output
Visualization	tcsBU / D3.js	Multi-trait concentric rings + legend, interactive layout
Interpretation	built-in pure computation	Quality/diversity, genetic distance/PCoA, network topology

3. Installation & Running

3.1 Release build (recommended for end users)

- Download the platform package from [NetST GitHub Releases](#).
- Extract and launch from **inside** the extracted directory (NetST.app on macOS, NetST.exe on Windows). No installation required.
- **Important:** run the program from within the extracted NetST folder. External dependencies (MAFFT, MUSCLE, fastHaN, McAN) are bundled and linked per platform.

Platform	Run from source	Bundled analysis programs	PyInstaller spec
macOS Apple Silicon	Yes	MAFFT, MUSCLE 3, fastHaN, RMST, McAN	netst-mac-arm64.spec
Windows x86-64	Yes	MAFFT, MUSCLE 3, fastHaN, RMST, McAN	netst-win.spec
Linux	Code path exists	Provide your own	Not provided

McAN is bundled for both supported platforms. NetST probes the executable and automatically selects the version-specific output option.

3.2 Run from source (developers / Linux users)

Use Python 3.10, 3.11, or 3.12 and an isolated virtual environment.

macOS / Linux

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
python main_form.py
```

Windows PowerShell

```
py -3.10 -m venv .venv
.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
python main_form.py
```

Main Python dependencies: PyQt6, PyQt6-WebEngine, and chardet (PyInstaller is only needed to build releases). RMST does not require NumPy. If the bundled MAFFT is unavailable, the program falls back to a system mafft on PATH.

3.3 Packaging

```
python -m pip install -r requirements-build.txt
python scripts/build.py
```

```
# Optional Windows one-file mode
python scripts/build.py --onefile
```

macOS produces dist/NetST.app; Windows produces the complete dist/NetST/ directory by default. The build driver checks dependencies, architecture, bundled tools, the packaged structure, and Qt WebEngine resources. Build natively on each target OS; signing and notarization remain maintainer release steps.

4. Interface Overview

The main window has a **menu bar**, a **tabbed workspace**, and a bottom **status bar**.

4.1 Menu bar

Menu	Items
File	Import and Replay Project; Export Project Configuration; Import FASTA / NEXUS / PHYLIP / VCF; Load Metadata; Export Sequence Data; Export Metadata; Exit
Analysis	Build / Rebuild Haplotype Network; Interpretation Analysis ► Sequence Quality and Diversity / Genetic Distance and PCoA /

Menu	Items
	Network Topology Metrics
Tools	Multiple Sequence Alignment (alignment only); Calculate Haplotype (align + haplotypes, no network); Sequence Format Conversion; Language (/ English)
Help	About; TCS-BU Help; NetST Help

4.2 Tabbed workspace

Tab	Content
Home	Software positioning, core capabilities, quick workflow, supported algorithms, and I/O overview
Network	Interactive tcsBU haplotype network (embedded Qt WebEngine)
Data	Sequence table: ID, name, sequence (read-only), analysis checkbox; names/selection/traits are editable
Metadata	Appears after metadata import: configure each trait's type, Group, visibility, and colour
Haplotype	Haplotype summary and sample-to-haplotype mapping
Alignment	Alignment matrix (renders the first 500 sites for responsiveness)
Interpretation	Structured interpretation results: overview KPI cards, chart visualizations, detail tables

In the tcsBU Network view, the Data (left) and Info (right) side panels can be collapsed and drag-resized to give the network more room.

4.3 Status bar

While an analysis runs, the bottom of the window shows a progress bar and a **Cancel** button. Cancelling requests the background task to stop and terminates the entire process group of the current external program; the same cleanup runs when the program closes or a new task replaces an old one.

5. Quick Start (Example Data)

Two SARS-CoV-2 example datasets are provided:

- **SL-SARS-CoV-2**: 130 aligned sequences with L / S lineage labels (categorical trait).
- **DP-SARS-CoV-2**: 482 genome sequences (72 from the Diamond Princess + 410 worldwide) with geographic location (categorical) and collection date (continuous).

Shortest path using DP-SARS-CoV-2:

1. **File** → **Import FASTA**, choose examples/DP-SARS-CoV-2/DP-SARS-CoV-2.fasta. Keep each complete FASTA header as the sample name in the standardization dialog.
2. **File** → **Load Metadata**, choose DP-SARS-CoV-2.meta.csv. Map ID to **Sample name**, Location to **Categorical trait + Group**, and Time to **Continuous trait**. For Time, use the date conversion so values such as 2019/12/26 become a numeric interval.
3. On the **Data** tab, verify that all names matched and select the samples to include (all are selected by default).
4. On the right, enter a **project name** and choose the **output directory** (default ~/HaplotypeOutput).
5. **Analysis** → **Build / Rebuild Haplotype Network**, choose a method and run. NetST performs alignment when needed, identifies haplotypes, constructs the network, and opens the visualization.
6. Inspect the **Network**, **Alignment**, and **Haplotype** tabs. Change displayed traits or colours on **Metadata**, then click **Apply Visualization Config** without rebuilding topology.
7. Run optional analyses under **Analysis** → **Interpretation Analysis**; results appear on the **Interpretation** tab.
8. Export tables from **File**, figures from **Save Image**, or a portable record from **File** → **Export Project Configuration**.

Composite FASTA headers such as sample_01|population_A|12.5 can still be split by delimiter and 0-based field position, but the supplied DP example uses a separate metadata file.

6. Data Import & Preprocessing

6.1 Supported sequence formats

Format	Menu	Notes
FASTA	File → Import FASTA	Whitespace in headers is normalized to underscores so PHYLIP/fastHaN don't split a name into fields
NEXUS	File → Import NEXUS	DNA/RNA MATRIX in DATA/CHARACTERS

Format	Menu	Notes
		blocks: sequential, interleaved, quoted names, nested comments, MATCHCHAR
PHYLIP	File → Import PHYLIP	Sequential and common interleaved; the declared count and length must match the data
VCF / VCF.GZ	File → Import VCF	See §6.3

The three non-VCF formats all enter the same **name-standardization dialog** and load into the Data tab as the current raw data. A failed import does not overwrite existing table data.

6.2 Standardization dialog (sample-ID handling)

The import dialog can:

- **clean / replace** characters in sample names;
- extract **sample name, categorical trait, continuous trait** from a composite header by **delimiter + 0-based field index** (compatible with legacy headers like sample_01|population_A|12.5);
- or process only the sample name and later import multiple traits via **File → Load Metadata**;
- optionally **filter sequences containing RYSWKMBDHDVN ambiguity codes**.

Continuous traits must be valid numbers.

6.3 VCF import

File → Import VCF accepts VCF / VCF.GZ with per-sample genotype columns. The sequence workflow targets a **single sequence region**, so the VCF must contain a single contig and each record must carry a GT field.

Metadata file (optional)

Provided as	Effect
None	metadata parsed from sample IDs only
Plain CSV/TSV	matched by sample name; fills categorical (group) and continuous traits
McAN six-column TSV	fills a categorical trait merged from Country/State/City; if sample

Provided as	Effect
	names are unchanged , enables native McAN VCF analysis and keeps real sampling dates for network orientation
McAN six-column format: SampleName AccessionID SamplingDate Country State City, missing values as *, header optional. If you rename/remove samples in standardization or later edit the table, the program automatically falls back to the aligned-FASTA → mutation adapter path.	
Alignment reconstruction rules	
• No reference FASTA: each VCF record becomes an equal-width variant-site block, producing a "variant-sites-only alignment". • With reference FASTA: each VCF REF is validated against the reference; invariant regions are filled back to produce a full-length alignment. • Multi-allelic sites resolve by GT allele number; heterozygous single-base sites use IUPAC codes, heterozygous complex indels use N. • Symbolic ALTs, breakends, multiple contigs, and mutually overlapping records are explicitly rejected.	

7. Metadata Management & Visualization Config

7.1 Importing metadata from CSV/TSV

File → Load Metadata imports a standalone metadata table. The first row is a header, and at least one column must map 1-to-1 to already-imported sequence names:

```
sample,population,value
sample_01,population_A,12.5
sample_02,population_B,13.1
```

The delimiter is auto-detected from the first data row (tab preferred). Each column can be set to **Sample name / Ignore / Categorical trait / Continuous trait**; you may select any number of traits at once, but **at least one categorical trait must be designated as the Group**. The program checks that sample names match completely on both sides and rejects duplicate names on either side, preventing silent overwrite or mismatch.

For every Continuous column, click **Continuous conversion** and choose one rule:

Input type	Accepted input and output
Plain number	Strict finite numeric values; blanks remain blank
Date / time	YYYY-MM-DD, YYYY/MM/DD, YYYY-MM, YYYY, or ISO date-time → elapsed days, calendar months, or

Input type	Accepted input and output
	calendar years from a selected start date; blank start means the earliest valid date
Measurement with unit	Length (mm/cm/m/km/in/ft), mass (mg/g/kg), or temperature (°C/°F/K) → one selected output unit; choose the assumed source unit for bare values

The preview checks the complete column, not only the visible first rows. Invalid input prevents confirmation and identifies its original metadata row. Conversion does not edit the source file; the normalized numeric result is stored in the current Data table and feeds the ordinary continuous visualization path. Calendar intervals are intuitive: the same day in the next month is exactly 1 month, and the same date in the next year is exactly 1 year; partial intervals remain fractional so the colour gradient stays continuous.

7.2 The Metadata tab

After import, a **Metadata** tab appears. Each row is one trait; you can switch categorical/continuous, choose the single Group, toggle visibility, and edit colours (categorical palette or continuous low/high endpoints, via a colour picker or a hex value). Categorical traits with 2–10 classes use curated class-count-specific palettes; larger sets receive stable, non-repeating generated colours. The default continuous gradient is gray (#BDBDBD) to black (#000 000). The program prevents deleting the last categorical Group.

Network nodes draw **concentric rings** in the order of the Metadata table: the **Group innermost**, then other enabled traits outward. A categorical ring is segmented by the class proportions of a haplotype's members; a continuous ring first sorts member values numerically from low to high, then draws the sectors in that order using the trait's range and gradient. The tcsBU **Legend** lists every class swatch and continuous gradient range in the same inner-to-outer order.

7.3 When to rebuild vs. only refresh the visualization config

Network **topology** is determined by the input samples and sequences; the metadata's types, grouping, rings, and colours only decide **how** existing topology is displayed.

Change	Action
Import a new FASTA/NEXUS/PHYLIP/VCF	Rebuild the haplotype network
Change sample names, sequences, or the analysis selection	Rebuild the haplotype network
Change the network algorithm/parameters	Rebuild the haplotype network
Import or edit metadata values	Only click Apply Visualization

Change	Action
	Config on the Metadata tab
Change trait type, Group, Visualize, or colour	Only click Apply Visualization Config
Adjust group names, colours, or sample-group membership in tcsBU	Only update the current visualization config
A lightweight refresh rewrites only tcsBU's _hapconf.csv, _groupconf.csv, _traitconf.csv, the embedded-data .js, and .html; the existing .gml, alignment, and sample-to-haplotype mapping are unchanged (identical sequences keep the same H1/H2... labels).	

8. Alignment & Haplotype Identification

8.1 Multiple-sequence alignment (MSA)

NetST aligns with **MAFFT** (several modes) or **MUSCLE**:

MAFFT mode	Character
Auto	automatic selection
FFT-NS-1	very fast, rough
FFT-NS-2	fast
G-INS-i	global, slower
L-INS-i	local, most accurate
E-INS-i	long unaligned regions
• Full network analysis: if input lengths differ, MAFFT runs first, falling back to MUSCLE on failure; equal lengths are treated as already aligned and go straight to haplotype processing. • Tools → Multiple Sequence Alignment: alignment only, for visual inspection or fine-tuning; most users do not need it separately.	

8.2 Haplotype identification

- **Tools → Calculate Haplotype:** align and compute haplotypes without building a network.
- Identical aligned sequences collapse into unique haplotypes (H1, H2, ...) to streamline the network. Results appear on the **Haplotype** tab and generate project_hap.fasta, project_seq.meta.csv, etc.

9. Haplotype Network Construction

Analysis → Build / Rebuild Haplotype Network opens the construction dialog. Pick an algorithm and parameters to run alignment → haplotype identification → construction → visualization in one step.

9.1 Algorithms & parameters

Algorithm	Engine ID	Parameters	Ideal for
Original TCS (statistical parsimony)	original_tcs	threads, ambiguous sites, merge intermediate nodes	intraspecific phylogeography, shallow divergence
Modified TCS	modified_tcs	threads	improved TCS variant
MSN (minimum spanning)	msn	epsilon	closely related, low-divergence populations
MJN (median-joining)	mjn	threads, epsilon	recombination/ missing data, ancestral inference
RMST (randomized minimum spanning tree)	rmst	exact/random mode, replicates, random seed, exclude ambiguous sites	bundled native engine, see §14.1
McAN (minimum-cost arborescence)	mcan	threads, reference sequence, exclude ambiguous sites	directed inclusion network, see §14.2

9.2 Choosing an algorithm

- **MSN**: simplest connectivity — intraspecific / population-level, low-divergence data.
 - **MJN**: accommodates reticulate evolution and infers ancestral haplotypes — complex reconstructions, ancient DNA, viral dynamics.
 - **TCS family**: statistical-parsimony representations commonly used for shallow, intraspecific variation; Original and Modified TCS expose different fastHaN options.
 - **RMST**: bundled native C++ engine; exact mode is deterministic and reproducible — a fast, robust default.
 - **McAN**: a mutation-inclusion network rooted at the reference; time-orientable when real sampling dates are provided.
-

10. Network Visualization (tcsBU)

The network renders on the **Network** tab with an enhanced **tcsBU** / **D3.js** view embedded in Qt WebEngine.

10.1 Multi-trait concentric rings

Each haplotype node is drawn as concentric rings: **Group innermost**, other enabled traits outward. Node size scales with frequency. The **Legend** lists every class swatch and continuous gradient range in inner-to-outer order.

10.2 Toolbar

- **Save Image**: export **SVG (vector)** / **PNG** / **JPG** / **PDF**, then choose a location in the system save dialog. SVG is standard image/svg+xml; PNG/JPG default to 2× resolution.
- **Zoom In** / **Zoom Out**, **Delete Node** / **Delete Link** (interactive editing).
- **Legend**: show/hide the legend.
- **Haplotype** / **Name-ID** / **Distance**: show the H1/H2 label, a selected member name/ID, or the per-edge changes count.
- **Edge Weight**: scale edge width inversely by the numeric changes count; **Undo** restores the latest deleted node or link.
- **Advanced**: open the advanced-settings dialog (below).

Use the right **Info** panel to search for a node, inspect node/edge attributes, highlight one item, control labels for one node or edge, and adjust per-node text position and size. See the full [tcsBU help](#) for standalone configuration files and all controls.

10.3 Advanced settings dialog

The dialog can be **moved by dragging its title bar** and is organized into three sections:

Force-Directed Layout Settings — Link Distance, Link Strength, Friction, Charge, Gravity, plus Start / Stop.

Node and Edge Settings

Parameter	Meaning	Default
Node Radius	radius (px) of a frequency-1 haplotype node; preserves relative sizes when adjusted	5
Node Line Width	node outline stroke width	0
Edge Line Width	base edge (link) line width	1
Edge Weight Scale	maximum Edge Line Width multiplier when Edge Weight is enabled	2
Text Offset	label offset beside a node	5
Haplotype Font Size	global H1/H2 label	13

Parameter	Meaning	Default
Name/ID Font Size	size; one node can override it in Info global member-name label size; one node can override it in Info	13

Metadata Ring Settings

Parameter	Meaning	Default
Ring Line Width	ring-segment stroke width	0.1
Ring Width Ratio	each non-Group ring width as a fraction of the frequency-scaled node radius	0.5
Outer Ring Ratios	comma-separated ratios inner→outer (missing values use 1)	—

Node/Edge Line Width are independent, separately adjustable parameters. Edge Weight reads the numeric Changes distance and scales every edge from the configured base line width; fewer changes produce thicker edges.

10.4 Read-only Data column

The **Sequence column on the Data tab is read-only** to prevent accidental edits from invalidating an existing alignment, haplotypes, and network. Sample names, selection state, and metadata traits remain editable.

11. Interpretation Analysis

Analysis → Interpretation Analysis offers three pure-computation analyses. The computation layer uses an **immutable aligned-sequence snapshot**, decoupled from the UI and file output, and runs on a background thread (cancellable). Results appear on the **Interpretation** tab and are saved as JSON.

Menu item	Input & computation	Main results
Sequence Quality and Population Genetics	current selected equal-length aligned sequences; complete or pairwise deletion; categorical population trait	S, Hd, π , θW , mean pairwise differences, Tajima's D, mismatch distribution, Hudson FST, one-level AMOVA/ Φ ST,

Menu item	Input & computation	Main results
Genetic Distance and PCoA	only A/C/G/T as called states; gap, N, ?, and IUPAC ambiguity pairwise-deleted	permutation P values p-distance matrix, comparable sites per pair, PCoA coordinates, positive/negative eigenvalue diagnostics
Network Topology Metrics	the current project GML, or a user-selected tcsBU-compatible GML	node/edge counts, connected components, density, cycle rank, degree, closeness, betweenness, articulation points, bridges

Before running, **Sequence Quality and Diversity** and **Genetic Distance and PCoA** show a parameter dialog: the former offers the missing-data policy, population trait, and permutation count for global FST/AMOVA; the latter sets the minimum comparable sites and minimum comparable coverage.

11.1 Result-tab visualizations

The Interpretation tab organizes results as **Overview + Charts + Detail tables** so you can grasp the data and network intuitively:

- **Overview:** a row of colour-coded **KPI cards** (e.g. missing rate graded green/amber/red), alongside a full metrics table and warnings/notes.
- **Charts (Visualizations):**
 - *Diversity:* group-diversity small multiples (N / haplotypes / H_d / π / θ_W), a per-sample missing-rate bar chart (coloured by threshold), and a **per-site variation-and-missing "track"** along the alignment (dark = parsimony-informative; bottom lane = missing-rate heat strip).
 - *Distance/PCoA:* a **PCoA ordination scatter** (coloured by group, with legend and explained-variance axis labels) and a **genetic-distance heatmap** (grey cells = too few comparable sites, distance unavailable).
 - *Topology:* a node-degree histogram and a hub-node betweenness ranking (**articulation points highlighted red**).
- **Detail tables:** retain all exact values for reference and cross-checking.

With PyQt6-WebEngine, charts render in a web view with hover tooltips; otherwise they fall back to static SVG (equally complete). Large tables limit displayed rows/columns for responsiveness; the complete result is always saved as JSON.

11.2 Missing-data & interpretation conventions

- RNA U is unified to T internally; -/N/?/IUPAC ambiguity are not treated as called alleles.

- Sequence pairs below the minimum comparable sites or coverage record distance as **missing** — never filled with 0 or an artificial saturation constant.
 - PCoA uses classical multidimensional scaling; it does not run PCA on the distance matrix as if it were ordinary features.
 - The direction of a McAN directed GML is kept as provenance; current centrality, articulation, and bridge metrics use an **undirected projection**.
 - **Topological centrality, bridging nodes, and PCoA proximity are exploratory descriptors — never automatically equated with ancestry, origin, transmission source, or true populations.**
 - **Tajima's D and mismatch distributions use population-complete sites.** Their biological interpretation still requires appropriate neutrality, recombination, sampling, and demographic assumptions.
-

12. Data Export & Format Conversion

12.1 Export

File → Export Sequence Data exports all records on the Data tab:

Target format	Notes
FASTA	header name discrete continuous, preserving both sequence and metadata
NEXUS / PHYLIP	writes name and sequence only (no metadata in the identifier); requires equal-length sequences
VCF	writes sequence variants and a NetSTSampleMetadata header, plus a matching _metadata.csv; requires equal length and at least one variant site

File → Export Metadata produces a standalone CSV/TSV metadata table with only sample and the traits on the current Metadata tab.

12.2 Sequence format conversion

Tools → Sequence Format Conversion:

Input	Output	Notes
FASTA	NEXUS, PHYLIP, VCF	NEXUS/PHYLIP require equal length; VCF can specify a reference sample
NEXUS	FASTA, PHYLIP, VCF	reads

Input	Output	Notes
		DATA/CHARACTERS matrix; sequential, interleaved, MATCHCHAR
PHYLIP	FASTA, NEXUS, VCF	sequential or common interleaved; declared count/length must be correct
VCF / VCF.GZ	FASTA, NEXUS, PHYLIP	reference FASTA optional; without it, a variant-sites alignment is produced

Converting to VCF requires equal-length aligned sequences; consecutive variant columns are merged into valid allele blocks and reference anchors are added for indels. A fully invariant alignment cannot produce a variants-only VCF. Input and output paths must differ.

13. Output Files

All generated files share the "output directory + project name" prefix (examples use project):

File	Content
project.fasta	raw input sequences for this run
project_aln.fasta	aligned sequences
project_seq.fasta / project_seq.phy	analysis sequences with original names / PHYLIP input for fastHaN
project_hap.fasta	de-duplicated H1, H2, ... haplotype sequences
project_seq.meta.csv	sample, haplotype, and trait mapping
project_hap_trait.csv / project_seq_trait.csv	haplotype summary / per-sample trait table
project_traitconf.csv	continuous-trait config (only when valid values exist)
project.gml	tcsBU-dialect network graph (fastHaN / RMST / McAN converted)
project_rmst.json / project_rmst.tsv	RMST parameters, sites, edge table, random-sampling stats (RMST only)

File	Content
project_mcan/	McAN input, sample-alias mapping, and original GraphML/JSON (McAN only)
project_hapconf.csv / project_groupconf.csv	tcsBU haplotype / group-and-colour config
project.js / project.html	local interactive visualization assets
project_diversity_analysis.json	sequence QC, overall/group diversity, and computation warnings
project_distance_analysis.json	p-distance, comparable sites, PCoA eigenvalues and coordinates
project_topology_analysis.json	network overview, node/edge topology metrics, direction provenance
project.netst.json	replayable project manifest: source paths/SHA-256, standardization, samples and traits, workflow parameters, tool/run provenance, output hashes, and final image view state

The Alignment and Haplotype tabs render only the first 500 sites for responsiveness; output files always keep the complete sequences.

Replaying a project

After sequence import, NetST updates project.netst.json in the output folder. Each imported sequence, VCF companion, reference, and metadata file is copied to inputs/<role>/ inside that project folder. **File** → **Export Project Configuration** saves the current manifest elsewhere and copies its managed inputs beside the new manifest. **File** → **Import and Replay Project** validates every referenced input against its saved SHA-256 (prompting to relocate missing files), restores the editable sample/trait state, reruns the recorded network and interpretation workflow, then applies the network state captured at image export. A changed source file is rejected instead of silently producing a different result. Manifest paths are relative to the .netst.json location, so transferring the complete project directory preserves replayability.

14. Methods & Algorithms

14.1 RMST built-in implementation

RMST (Randomized Minimum Spanning Tree) is implemented by the bundled dependency-free C++17 netst-rmst executable. It reads project_hap.fasta and project_seq.meta.csv and

computes uncorrected mutation counts (Hamming distance) over unique haplotypes. Two modes:

- **Exact mode (default, recommended):** by distance layer, determines all edges that can appear in at least one minimum spanning tree; deterministic and seed-independent.
- **Random mode:** repeatedly randomizes haplotype order and runs stable Kruskal, reporting each edge's occurrence count and frequency; a fixed seed is reproducible, but finite replicates do not guarantee finding every compatible edge.

The native random mode uses a platform-independent SplitMix64 permutation stream. A seed is stable between the new macOS and Windows binaries, but it does not reproduce the historical NumPy RNG stream byte-for-byte.

By default any alignment column containing a character outside A/C/G/T/U/- is excluded, RNA U is unified to T, and gap is a comparable state. Haplotypes that become identical after filtering are merged with their sample members into one node, recorded in the JSON warnings and node haplotypes fields. Scale limits: exact mode ≤ 1000 filtered nodes; random mode ≤ 500 nodes and ≤ 1000 replicates.

Reference: Paradis, E. (2018) *Analysis of haplotype networks: The randomized minimum spanning tree method*. Methods in Ecology and Evolution 9:1308–1317.

14.2 McAN adapter

McAN natively accepts a VCF or mutation + metadata + site mask and outputs GraphML/JSON. `service/mcan_adapter.py` supports two paths:

- **Native VCF path:** importing a VCF with McAN six-column metadata and unchanged sample names calls McAN `--vcf` directly, keeping real sampling dates for network orientation.
- **Mutation adapter path:** FASTA/PHYLIP sources, VCFs without metadata, or samples renamed/edited:
 1. read the aligned FASTA and compute per-site differences against the chosen reference;
 2. generate mutation/metadata with internal aliases like S0000001 to avoid GraphML corruption from unescaped names;
 3. produce an explicit site mask (optionally excluding columns with non-A/C/G/T/U/gap characters);
 4. call McAN in a dedicated `project_mcan/` directory, preserving the original GraphML/JSON;
 5. convert to a tcsBU-readable GML, restoring original names and network distances.

NetST limits the maximum McAN sequence coordinate to 30000 and rejects longer alignments before calling it. The adapter automatically selects McAN 1.2 `--outDir` or McAN 1.4.x `--out`, and supports both GraphML formats. When a FASTA has no sampling dates, it does not fabricate a time order — it yields a mutation-

inclusion network rooted at the reference. Reference: Li L, et al. (2023) *McAN: a novel computational algorithm and platform for constructing and visualizing haplotype networks*. Briefings in Bioinformatics 24:bbad174.
<https://doi.org/10.1093/bib/bbad174>

14.3 Interpretation metric definitions

- **Sequence quality:** per-sample/per-site missing, gap, unknown, and ambiguity counts and rates; effective sites, variable sites, parsimony-informative sites.
 - **Diversity:** haplotype richness, haplotype diversity H_d , nucleotide diversity π , Watterson's θ_W , segregating sites S , group private haplotypes.
 - **Genetic distance:** pairwise-deletion p-distance, reporting the effective comparable sites for each pair.
 - **PCoA:** classical multidimensional scaling (the dependency-free solver defaults to ≤ 200 sequences; larger sets still output the distance matrix but skip PCoA with a warning).
 - **Network topology:** connected components, density, cycle rank (independent loops), degree, closeness, betweenness, articulation points, bridges. Mutation count is always treated as distance, never as stronger connectivity.
-

15. FAQ & Known Limitations

Q: Which population-structure estimators are implemented? A: NetST reports equal-population Hudson sequence FST and one-level AMOVA/ Φ ST with optional label permutations. It does not yet provide hierarchical AMOVA, genotype-based Weir-Cockerham FST, or demographic-model fitting for mismatch distributions.

Q: I edited metadata or colours — do I need to rebuild the network? A: No. As long as sequences and the sample set are unchanged, click **Apply Visualization Config** on the Metadata tab (see §7.3).

Q: The network view is blank / text only? A: tcsBU depends on Qt WebEngine. Without PyQt6-WebEngine, the network tab falls back to a degraded text widget; install that dependency.

Known limitations

- No complete Linux release yet; Linux needs compatible external programs supplied by the user.
- Bundled external binaries are platform-specific and cannot be used across architectures. Windows ships static-runtime x86-64 McAN and RMST executables; Linux engines must still be supplied separately.
- The McAN adapter accepts at most 30000 alignment sites.
- Native RMST: exact mode ≤ 1000 nodes and random mode ≤ 500 nodes; randomized work is additionally bounded at 50 million edge evaluations.

- VCF sequence conversion is limited to single-contig, non-overlapping small variant records; no structural variants, breakends, or symbolic ALTs.
 - The dependency-free PCoA solver defaults to ≤ 200 sequences.
 - Full GUI interaction, platform packaging, and real-data end-to-end workflows need manual verification on target systems.
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16. Citation & Acknowledgements

The manuscript citation is provisional until journal publication:

Zhang Z, Song Y, Yu X, Hou J, Yu Y. *NetST: a desktop application for project-based haplotype network analysis with metadata integration*. Manuscript.

Please also cite the relevant dependencies:

- Chi L, Zhang X, Xue Y, Chen H. 2025. *fastHaN: a fast and scalable program for constructing haplotype network for large-sample sequences*. Molecular Ecology Resources 25:e13829. <https://doi.org/10.1111/1755-0998.13829>
- Múrias Dos Santos A, et al. 2016. *tcsBU: a tool to extend TCS network layout and visualization*. Bioinformatics 32:627–628. <https://doi.org/10.1093/bioinformatics/btv636>
- Nakamura T, et al. 2018. *Parallelization of MAFFT for large-scale multiple sequence alignments*. Bioinformatics 34:2490–2492. <https://doi.org/10.1093/bioinformatics/bty121>
- Paradis E. 2018. *Analysis of haplotype networks: The randomized minimum spanning tree method*. Methods in Ecology and Evolution 9:1308–1317. <https://doi.org/10.1111/2041-210X.12969>
- Li L, et al. 2023. *McAN: a novel computational algorithm and platform for constructing and visualizing haplotype networks*. Briefings in Bioinformatics 24:bbad174. <https://doi.org/10.1093/bib/bbad174>

Example-data citation:

- DP-SARS-CoV-2: Sekizuka T, et al. 2020. *Haplotype networks of SARS-CoV-2 infections in the Diamond Princess cruise ship outbreak*. PNAS 117:20198–20201.
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17. Contact

For any questions, suggestions, or comments about NetST, please contact: yyu@scu.edu.cn
• zzhen0302@163.com

This manual is based on the current NetST source and README and may change as the software evolves. License: MIT.